

AXIOM SYSTEM PROVISIONING SPECIFICATION

HEADLESS MULTI-AGENT ECOSYSTEM LAYOUT & WEB FRAMEWORK FOR CLAUDE AGENT BLUEPRINT

Operational Objective: Build the software cluster from scratch with 99.9% state resilience, self-repair mechanisms, and a 7-office professional portal wireframe mapping human personas into specialized computing runtimes.

1. Executive Operational Protocol

This master document is meticulously optimized to present code bases, network routing layers, automated subscription models, and UI mapping logic in the most compact layout possible. It is explicitly tailored for prompt-injection feeding into LLM developer agents like **Claude**. Read sequentially to orchestrate database configurations, isolated agent processes, dynamic financial fee absorption calculations, and the front-facing web portal layout.

2. Core Ecosystem & Price Absorption Architecture

To insulate platform margins from third-party transaction expenses, every consumer charge is programmatically wrapped via the formula: **Final = (Target_Price + 0.30) / (1 - 0.029)**. This converts baseline operational fee structures directly into dynamic API payloads before submission to Stripe.

Service ID	SaaS Operational Layer / Office Module	Base Target	Stripe Formula Applied	API Payload	Compute Architecture Tier
subscription	SaaS Platform Base Access Module	\$99.00 USD	$(99.00 + 0.30) / 0.971$	10227 Cents	Shared CPU / Persistent DB
estimate	Office 1: AI Automotive Damage Estimator	\$5.00 USD	$(5.00 + 0.30) / 0.971$	546 Cents	NVIDIA Cluster / Vulkan VRAM
claims	Office 2: Automated Insurance Claims Desk	\$5.00 USD	$(5.00 + 0.30) / 0.971$	546 Cents	Vision LLM API Proxy Node
total_loss	Office 3: Total Loss Mathematical Actuary	\$5.00 USD	$(5.00 + 0.30) / 0.971$	546 Cents	High-Compute Analytics Node
lien	Office 4: Mechanic's Lien & Legal Recourse Bureau	\$99.00 USD	$(99.00 + 0.30) / 0.971$	10227 Cents	State DMV REST Integration
audit	Office 5: Repair Order Invoice Audit Chamber	\$5.00 USD	$(5.00 + 0.30) / 0.971$	546 Cents	Contextual OCR Parsing Node
cpo	Office 6: Certified Pre-Owned Inspection Desk	\$5.00 USD	$(5.00 + 0.30) / 0.971$	546 Cents	Structured Schema Checker
gpu_resell	Office 7: Idle Infrastructure VRAM Brokerage	Dynamic %	Platform Allocation	Variable	Headless Mesh Brokerage Cluster

3. Automated Platform Code Blueprint

3.1 Environment Dependency Map (`pyproject.toml`)

```
[build-system]
requires = ["setuptools>=68", "wheel"]
build-backend = "setuptools.backends.legacy:build"

[project]
name = "AXIOM-ESTIMATE"
version = "1.1.0"
description = "Autonomous multi-agent SaaS cluster with absolute 99.9% self-healing high-availability"
requires-python = ">=3.11"
dependencies = [
```

```
"fastapi">=0.115", "uvicorn[standard]>=0.34", "pydantic">=2.11", "openai">=1.72",
"celery[redis]>=5.5", "redis">=5.2", "sqlalchemy">=2.0", "psutil">=7.0",
"stripe">=11.6", "prometheus-client">=0.21", "python-dotenv">=1.1", "httpx">=0.28"
```

```
]
```

3.2 Relational Multi-Tenant Storage Layer (`db/schema.sql`)

```
CREATE TABLE IF NOT EXISTS shops (
    id SERIAL PRIMARY KEY, name VARCHAR(255) NOT NULL, stripe_customer_id VARCHAR(100) UNIQUE,
    subscription_status VARCHAR(50) DEFAULT 'active', created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
CREATE TABLE IF NOT EXISTS jobs (
    id SERIAL PRIMARY KEY, shop_id INTEGER REFERENCES shops(id), service_type VARCHAR(50) NOT NULL,
    status VARCHAR(50) DEFAULT 'pending', vram_usage_bytes BIGINT DEFAULT 0,
    execution_time_ms INTEGER DEFAULT 0, retries_count INTEGER DEFAULT 0, created_at TIMESTAMP DEFAULT
CURRENT_TIMESTAMP
);
CREATE TABLE IF NOT EXISTS api_costs (
    id SERIAL PRIMARY KEY, job_id INTEGER REFERENCES jobs(id), gpu_compute_cost NUMERIC(10, 5) NOT NULL,
    llm_tokens_cost NUMERIC(10, 5) NOT NULL, total_platform_cost NUMERIC(10, 5) NOT NULL, billed_to_client
BOOLEAN DEFAULT FALSE
);
CREATE TABLE IF NOT EXISTS liens (
    id SERIAL PRIMARY KEY, shop_id INTEGER REFERENCES shops(id), job_id INTEGER REFERENCES jobs(id),
    vin VARCHAR(17) NOT NULL, amount_owed NUMERIC(10, 2) NOT NULL, lien_status VARCHAR(50) DEFAULT
'draft', created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

3.3 Mathematical Billing Controller (`axiom/billing.py`)

```
import stripe, os
from decimal import Decimal
stripe.api_key = os.getenv("STRIPE_SECRET_KEY")

class BillingEngine:
    @staticmethod
    def calculate_stripe_absorption(target_amount: float) -> int:
        target = Decimal(str(target_amount))
        fixed_charge = Decimal("0.30")
        stripe_percentage = Decimal("0.029")
        final_charge = (target + fixed_charge) / (1 - stripe_percentage)
        return int(round(final_charge, 2) * 100)

    @classmethod
    def collect_transactional_fee(cls, shop_stripe_id: str, service_type: str):
        base_price = 99.00 if service_type == "lien" else 5.00
        amount_in_cents = cls.calculate_stripe_absorption(base_price)
        return stripe.Charge.create(
            amount=amount_in_cents, currency="usd", customer=shop_stripe_id,
            description=f"AXIOM Charge: Tier [{service_type.upper()}]"
        )
```

3.4 Container Environment Core (`docker/Dockerfile.gpu`)

```
FROM nvidia/vulkan:1.3-rt-ubuntu22.04
ENV DEBIAN_FRONTEND=noninteractive
WORKDIR /app
RUN apt-get update && apt-get install -y python3.11 python3-pip vulkan-tools libvulkan-dev git && rm -rf /
var/lib/apt/lists/*
COPY pyproject.toml .
RUN pip3 install --no-cache-dir .
COPY . .
```

4. The 99.9% Autonomy & Automated Self-Healing Watchdog

To reach continuous uptime thresholds without human oversight, a localized watchdog script (`axiom/self\_healing.py`) operates concurrently to trap memory anomalies, trace message queue deadlocks, and explicitly recycle context loops:

```
import psutil, os, subprocess, logging

class SelfHealingEngine:
    @staticmethod
    def check_memory_threshold(max_vram_pct: float = 90.0):
        vram = psutil.virtual_memory()
        if vram.percent > max_vram_pct:
            logging.warning(f"VRAM leakage trapped: {vram.percent}%. Flushing task containers...")
            os.system("celery -A axiom.agent_core worker --loglevel=info --concurrency=4 &")

    @staticmethod
    def inspect_worker_queues():
        result = subprocess.run(["celery", "-A", "axiom.agent_core", "inspect", "ping"],
            capture_output=True, text=True)
        if "OK" not in result.stdout:
            logging.error("Asynchronous Agent cluster stalled. Executing hardware sub-process recycle...")
            subprocess.Popen(["celery", "-A", "axiom.agent_core", "worker", "--loglevel=info"])
```

5. Front-End Interface Strategy: The 7 Physical Office Persona Virtualization

To provide high trust to enterprise users, the commercial portal completely visualizes the 7 asynchronous code pathways as ****7 Distinct Professional Physical Offices****. The interface incorporates highly detailed structural wireframes modeling physical workers, real human environments, and crisp digital console arrays.

Office Tier 1: Immediate Damage Assessment Command Center

\$5.46 Total

Visual Asset Layout: Portrait of a senior automotive structural master wearing a dark charcoal inspection uniform, consulting a rugged industrial diagnostic pad inside a high-end cleanroom mechanic garage.

Autonomous Logic: Receives structural high-resolution uploads. The agent processes the file through a CUDA-accelerated vision array to locate physical impacts and generate cross-referenced estimates instantly.

UI Component: [Uploader Terminal] -> [Live VRAM Frame Buffer Stream] -> [Download Estimates PDF]

Office Tier 2: Automated Insurance Claims Negotiation Desk

\$5.46 Total

Visual Asset Layout: Close-up shot of an insurance mitigation officer in a bright corporate boardroom, analyzing multi-channel ledger displays on a multi-monitor display array.

Autonomous Logic: Extracts repair configurations and reformats them into standard CCC ONE / Mitchell compliance strings. It evaluates carrier guidelines to eliminate manually driven negotiations.

UI Component: [Carrier Dropdown Selector] -> [Asynchronous Claim Payload Logs] -> [Stripe Invoice Receipt]

Office Tier 3: Mathematical Total Loss Calculation Lab

\$5.46 Total

Visual Asset Layout: A data analyst looking intently at a matrix of glowing financial line charts, tracking real-time automotive salvage auction fluctuations.

Autonomous Logic: Connects directly to localized salvage databases and historical market tracking nodes. It runs automated matrix equations to determine the precise threshold when a vehicle must be labeled as a total loss.

UI Component: [VIN Entry Input] -> [Real-Time Salvage Threshold Matrix Gauge] -> [Total Loss Determination Vector]

Office Tier 4: Certified Mechanic's Lien Recovery Bureau

\$102.27 Total

Visual Asset Layout: Cinematic portrait of a legal processing specialist reviewing title documentation inside an office overlooking Miami, next to architectural blue prints and official stamps.

Autonomous Logic: Monitored by the state legislative legal agent. Upon unpaid service flags, the agent automatically scrapes registered vehicle titles, pulls certified bank address records, and files formal DMV mechanics lien paperwork.

UI Component: [Unpaid Balance Form] -> [Lienholder Bank Address Fetcher] -> [DMV Certified Mail Dispatch Tracer]

Office Tier 5: Forensic Invoice & Repair Order Audit Chamber

\$5.46 Total

Visual Asset Layout: Macro shot of a financial compliance investigator checking lines of data displayed on an ultra-wide clean monitor workspace.

Autonomous Logic: Runs rigorous background token scans comparing final parts list receipts against local parts catalogs. It flags unperformed tasks, inflated margins, or parts markups instantly.

UI Component: [Raw Invoice Dump] -> [OCR Discrepancy Stream] -> [Calculated Shop Profit Leak Gauge]

Office Tier 6: Autonomous Certified Pre-Owned (CPO) Inspection Board

\$5.46 Total

Visual Asset Layout: Sharp shot of a technical inspector checking the alignment sensors and diagnostic ports of a luxury vehicle using wireless scanners.

Autonomous Logic: Evaluates digital vehicle histories, onboard module logs, and shop records against strict manufacture multi-point inspection rules to issue instant digital CPO badges.

UI Component: [OBD-II Data Stream Intake] -> [Manufacturer Checklist Validator] -> [CPO Asset Encryption Anchor]

Office Tier 7: High-Compute Hardware Node & Distributed VRAM Brokerage

Dynamic yield

Visual Asset Layout: Symmetrical wide shot of an enterprise edge computing server stack lit with blue LEDs, housing arrays of liquid-cooled high-end processing units.

Autonomous Logic: Monitors the system's local memory load. When processing queues drop, the agent automatically repackages the unused hardware VRAM power and sells it back into global decentralized compute markets.

UI Component: [Idle VRAM Tracker] -> [Active Token Lease Metrics] -> [Automated Payout Balance Tracker]

6. Unified Clustered Single-Command Deploy Shell

6.1 Orchestration Infrastructure (`docker-compose.yml`)

```
version: '3.8'
services:
  db:
    image: postgres:15-alpine
    environment: { POSTGRES_DB: axiom_db, POSTGRES_PASSWORD: secret_secure_pass }
    volumes: [ "postgres_data:/var/lib/postgresql/data", "./db/schema.sql:/docker-entrypoint-initdb.d/
schema.sql" ]
    ports: [ "5432:5432" ]
  redis:
    image: redis:7-alpine
    ports: [ "6379:6379" ]
  api:
    build: { context: ., dockerfile: docker/Dockerfile.gpu }
    command: uvicorn api.main:app --host 0.0.0.0 --port 8000
    ports: [ "8000:8000" ]
    depends_on: [ db, redis ]
    environment: [ "DATABASE_URL=postgresql://postgres:secret_secure_pass@db:5432/axiom_db" ]
  celery_worker:
    build: { context: ., dockerfile: docker/Dockerfile.gpu }
    command: celery -A axiom.agent_core worker --loglevel=info --concurrency=4
    depends_on: [ redis, db ]
    deploy:
      resources:
        reservations:
          devices: [{ driver: nvidia, count: all, capabilities: [gpu] }]
volumes:
  postgres_data:
```

6.2 Single-Command Production Activation

Execute this unified bash instruction in your target instance root directory to pull base driver images, construct application layouts, instantiate background processes, and run the complete microservice network simultaneously:

```
docker compose up --build -d
```